

Raj Kamal Singh

Data Scientist | ML Engineer | AI Practitioner

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SUMMARY

Data Scientist and ML Engineer with 4 plus years of experience spanning statistical analysis, machine learning, GenAI systems, and production data pipelines. Currently at Fidelity Investments building automated ML-powered analytics that improve reporting accuracy and operational efficiency at scale. Proficient in Python, SQL, and the modern data stack with hands-on experience in deep learning, LLM fine-tuning, RAG pipelines, and cloud infrastructure on AWS. Published researcher in generative AI (IEEE 2022). M.S. in Data Science, University of Maryland (GPA 3.85). Comfortable moving between building forecasting models, fine-tuning LLMs, designing ETL pipelines, and presenting findings to senior leadership.

EDUCATION

M.S. in Data Science | University of Maryland, College Park, USA | May 2026

GPA: 3.86 | Coursework: Machine Learning, NLP, Statistical Modeling, Data Science at Scale, Probability and Statistics

B.Tech in Computer Science & Engineering | University of Petroleum & Energy Studies, India | Jun 2020

PUBLICATION

GAN and IEC Approach for Image Generation | IEEE ISMSIT 2022

- Trained and evaluated GANs for synthetic visual data generation; applied gradient penalty and spectral normalization to reduce mode collapse by 20%; conducted rigorous quantitative evaluation with full experimental documentation.

TECHNICAL SKILLS

Languages: Python (expert), SQL (advanced), R (familiar), C++

Data Science and ML: Scikit-Learn, PyTorch, TensorFlow, Pandas, NumPy, SciPy, Statsmodels, Regression, Classification, Time-Series Forecasting (LSTM), Anomaly Detection, Feature Engineering, A/B Testing, Hypothesis Testing, EDA

GenAI and LLMs: LLM Fine-tuning (FLAN-T5), RAG Pipelines, Prompt Engineering, Agentic AI, Hugging Face, LoRA, GANs, CNNs (ResNet-50), OpenCV, Claude

Cloud and Data Engineering: AWS (S3, EC2, Kinesis, Athena, Lambda, SNS), Snowflake, Redshift, Databricks, Apache Flink, Docker, FastAPI, ETL Pipelines, Git, CI/CD

BI and Visualization: Power BI (DAX), Tableau, Looker, AWS QuickSight, Matplotlib, Seaborn, Plotly, Advanced Excel (Power Query, Pivot Tables)

WORK EXPERIENCE

Data Analyst | Fidelity Investments, USA | May 2025 – Present

- Build Python and SQL-based ML workflows to process and validate large-scale financial datasets, improving reporting accuracy by 25% and reducing manual analysis effort by 40%; develop automated data quality monitoring pipelines with anomaly detection, cutting data inconsistencies by 30%.
- Design interactive KPI dashboards using Power BI, AWS Athena, and QuickSight, reducing stakeholder reporting turnaround by 50%; collaborate with data engineering teams to optimize ETL workflows and reduce data retrieval latency by 35%.
- Translate complex financial and operational data into executive-ready insights, communicating findings clearly to both technical and non-technical audiences across the organization.

Data Analyst | Nimbbi, Gurugram, India | July 2022 – July 2024

- Analyzed high-volume production datasets using SQL and Python, identifying failure patterns through root-cause analysis and hypothesis testing that drove a 30% reduction in production defects; built automated validation frameworks reducing manual verification effort by 45%.
- Partnered with product and engineering teams to develop KPI dashboards, translate analytical findings into engineering recommendations, and deliver data-driven insights to both technical and non-technical leadership.

Quality Analyst | Larsen and Toubro Infotech (LTI), Pune, India | June 2020 – June 2022

- Developed data-driven validation frameworks across 2,000 plus test scenarios, reducing post-release defects by 25%; automated test workflows using SQL and Python, improving efficiency by 35% and contributing to 97% on-time delivery.

PROJECTS

Stock Price Prediction Using LSTM | Python, TensorFlow, LSTM, FastAPI, Docker, AWS EC2 | 2025

- Built an end-to-end deep learning forecasting pipeline fusing technical indicators (RSI, MACD, SMA) and NLP-derived sentiment, achieving 15% accuracy improvement over baseline (MAE: 1.73, RMSE: 2.91); deployed as a production REST API on AWS EC2 with automated pipeline refresh and monitoring.

RAG-Based QA System with LLM Fine-Tuning | Python, Hugging Face, FLAN-T5, RAG, LoRA, Gradio | 2025–2026

- Scaled a generative QA pipeline from 20 samples to 87,000 plus examples (4,000 times increase), fine-tuning FLAN-T5 (80M to 250M parameters) with mixed-precision training and LoRA; built a RAG pipeline with embedding-based retrieval and deployed an interactive Gradio inference application end-to-end.

Real-Time Bitcoin Anomaly Detection | Python, AWS Kinesis, Apache Flink, Lambda, Isolation Forest, QuickSight | 2025

- Architected a production streaming pipeline on AWS (Kinesis, Flink, Lambda) processing 10,000 plus records per day at sub-second latency; trained and deployed an Isolation Forest model with SNS-based alerting, improving detection precision approximately 30%; built a QuickSight dashboard reducing manual monitoring effort by 60%.

Food Context Classification (Computer Vision) | Python, PyTorch, ResNet-50, OpenCV, Transfer Learning | 2025–2026

- Built an image classification pipeline using ResNet-50 transfer learning to classify food images under challenging real-world conditions; designed a 4-condition controlled ablation study achieving 94.52% test accuracy with confusion matrix analysis to identify failure modes.